

Maths & Science Olympiad Challenge



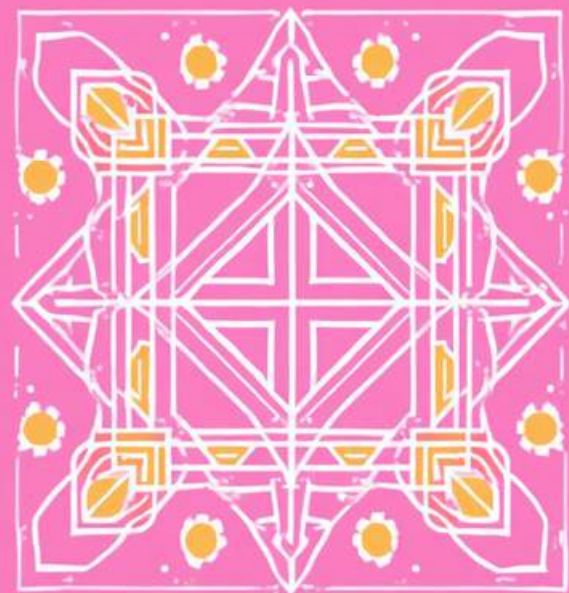


Hook & Mystery

Rangoli & Ghee Secrets

Can you crack the missing pieces in the Kolam?
Why does ghee change in winter?
Let's start our adventure as pattern
detectives and science explorers!

Decoding Patterns: Rangoli & Rotations



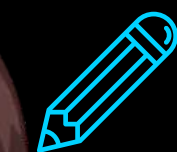
Shape Changes

Watch how shapes transform! Triangles become squares, circles grow bigger.



Rotation Rules

Triangle rotates 90° clockwise each step. Track the direction!



Shading & Dots

Dots increase by 2 each time. Count carefully to predict the next!



Pattern spotted! Rotate. Count. Predict.

Arithmetic Patterns:

The Indian Coins Puzzle



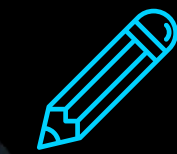
Different Ways

How many combinations can make ₹50?
Let's explore all the possibilities!



Counting Totals

Add up your coins: $10 \times ₹5 = ₹50$, or mix
₹5 and ₹2 coins together!

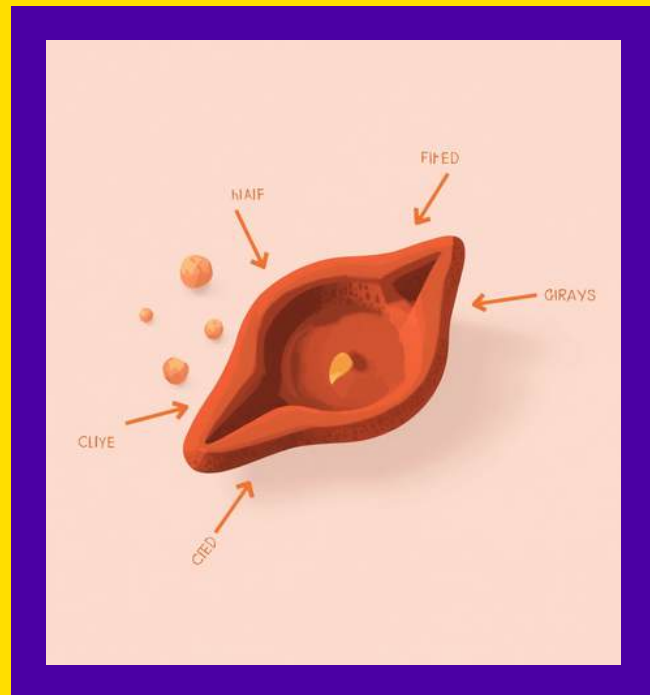
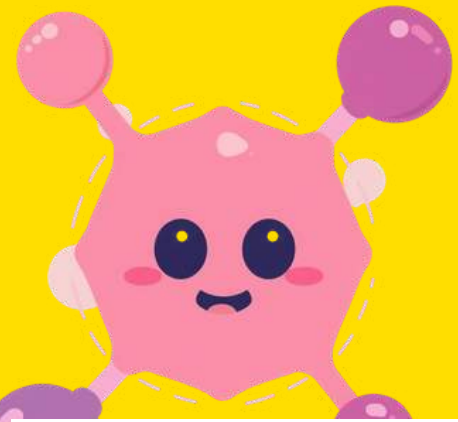


Systematic Approach

Start with max ₹5 coins, then swap for ₹2
coins step by step. Pattern spotted!



Molecular Detectives at Home



Feel the science in your home!



Solid & Liquid

Molecules: Tawa & Bowl

Solids hold shape, liquids flow and fill containers. Iron Tawa molecules are tightly packed and vibrate in place. Steel Bowl with water has loosely packed molecules that flow freely!

Gas Molecules: Agarbatti Aroma Magic

Gas particles move fast and fill the air!
When you light an agarbatti, the smoke
spreads everywhere in the room.
No shape, no fixed volume!
Particles zoom freely in all directions.





Shep Seating Matrix



Logical Reasoning

Five Friends at the Bus Stop

Meet Arjun, Bhavana, Charan, Diya & Esha in Bangalore!
Let's use logic to find who is tallest!

Logical Reasoning Strategy

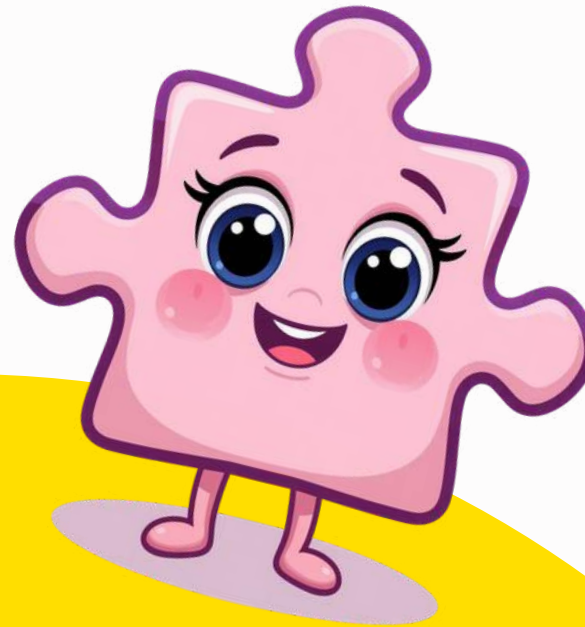
- Step 1: Read all clues carefully first
- Step 2: Start with the most definite clue
- Step 3: Eliminate impossible options one by one
- Step 4: Check your answer against all clues

Use a simple chart or ladder to track what you know. Cross out wrong answers as you go. Excellent deduction! Keep going!

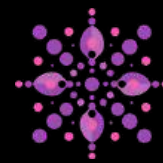


Guided Sprint: Olympiad Hack Challenge

Can you solve this multi-step puzzle?
Combine patterns, particles, and logic!
Strategy tip: Use elimination and look
for clues carefully in each step.

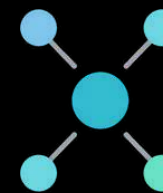


Session Wrap-Up & Mission



Pattern Analysis

Spot sequences, predict what comes next, and decode visual patterns like a pro!



Particle Properties

Understand how solids, liquids, and gases behave at the molecular level.



Logical Elimination

Use clues and systematic thinking to solve tricky puzzles step by step.

Assessment

Overview: Three Tiers



Tier 1: Conceptual

5 questions testing your foundational understanding of patterns, particles, and logic basics.



Tier 2: Application

5 questions requiring you to apply concepts to solve real-world problems.



Tier 3: Grandmaster

5 HOTS questions challenging your higher-order thinking and problem-solving skills!

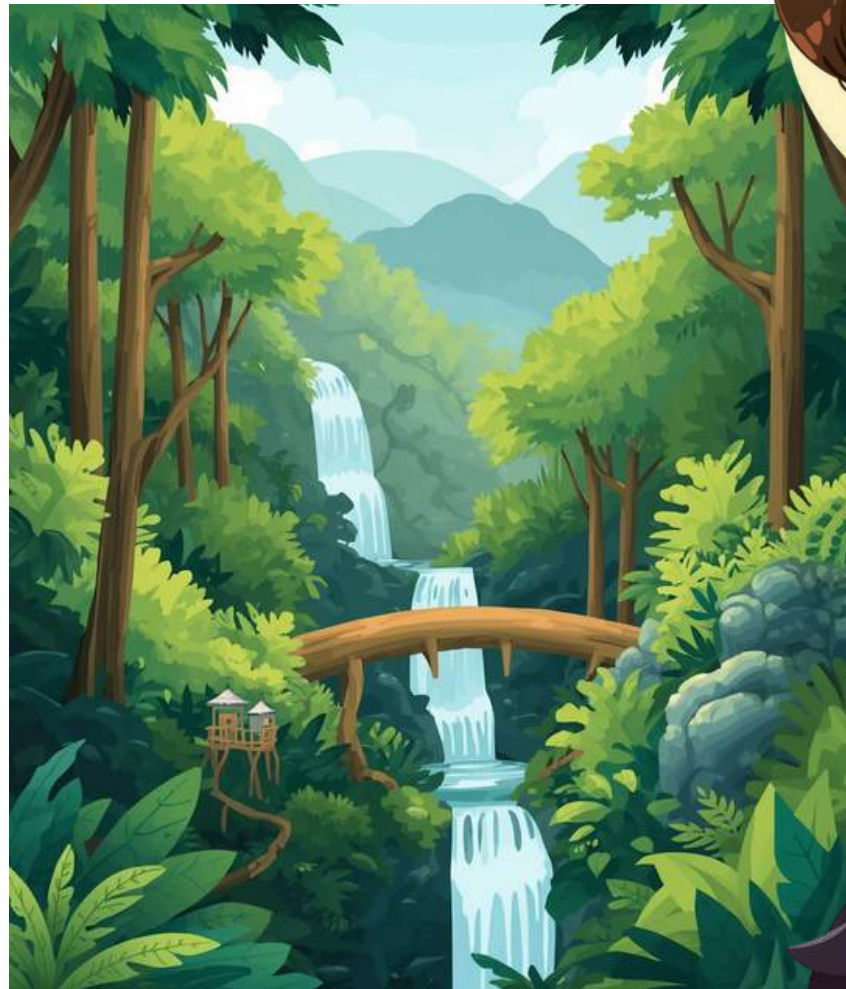
Indian Cultural Anchors in Learning

**Locations: Delhi Metro, Cherrapunji rainforests,
and Khari Baoli spice market!**

Objects: Rangoli, Tawa, Diya, Agarbatti

Names: Arjun, Bhavana, Charan, Diya, Esha

Our culture makes learning fun and meaningful!





Thank You!

Let's Crack the Code!

Ready for your next Olympiad adventure?
You are now pattern detectives and science explorers!

